

# SAMUEL A. BALLAS

Department of Mathematics  
208 Love Building  
1017 Academic Way  
Tallahassee, FL 32306-4510

sballas@fsu.edu  
<https://math.fsu.edu/~ballas>

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## Employment

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|--------------------------------------------------|---------------------------|
| <b>Florida State University</b>                  | Tallahassee, Florida      |
| • Director of Pure Mathematics                   | 2023-Present              |
| • Associate Professor                            | 2022-Present              |
| • Assistant Professor                            | 2016-2022                 |
| <b>University of California at Santa Barbara</b> | Santa Barbara, California |
| • RTG Visiting Assistant Professor               | July 2013-July 2016       |

## Education

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|--------------------------------------------------------------------------------------------|------------------|
| <b>University of Texas at Austin</b>                                                       | Austin, Texas    |
| • Ph.D., Mathematics                                                                       | May 2013         |
| Thesis: <i>Flexibility and Rigidity of Three-Dimensional Convex Projective Structures.</i> |                  |
| Advisor: Alan Reid                                                                         |                  |
| • M.A., Mathematics                                                                        | May 2011         |
| <b>Emory University</b>                                                                    | Atlanta, Georgia |
| • B.S. Mathematics., <i>Summa cum Laude</i>                                                | May 2007         |
| Thesis: <i>Geometrically Finite Fuchsian Groups and the Patterson-Sullivan Measure.</i>    |                  |
| Advisor: David Borthwick                                                                   |                  |

## Research Interests

Low-dimensional topology, hyperbolic geometry, real projective geometry, discrete groups, deformation theory, higher Teichmüller theory.

## Publications and Preprints

- *Parabolic-preserving deformations of cusped hyperbolic lattices* (with J. Paupert and P. Will), arXiv:2605.03161
- *Geometric Perspective on Concentration Phenomena in Frame Theory* (joint with F. Karabatsos and T. Needham), arXiv:2605.03867
- *On the existence of Parseval frames for vector bundles* (with T. Needham and C. Shonkwiler), Trans. Amer. Math. Soc. Ser. B 12 (2025), 395-416
- *Tame and relatively elliptic  $\mathbb{CP}^1$ -structures on the thrice punctured sphere* (with P. Bowers, A. Casella, and L. Ruffoni), Alg. Geom. Top. 24-8 (2024), 4589-4650
- *The moduli space of marked generalized cusps in real projective manifolds* (with D. Cooper and A. Leitner), Conf. Geom. & Dyn. vol 26 (2022), 111-164
- *Gluing equations for real projective structures on 3-manifolds* (with A. Casella), Geom. Dedicata 215 (2021) 69-131.

- *Thin subgroups isomorphic to Gromov–Piatetski-Shapiro lattices*  
Pacific J. Math. 309 (2020), no 2, 257–266.
- *Constructing thin subgroups of  $\mathrm{SL}(n+1, \mathbb{R})$  via bending*  
(with D.D. Long), Algebr. Geom. Topol. 20-4 (2020), 2071–2093
- *Constructing convex projective 3-manifolds with generalized cusps*  
J. London. Math. Soc. 103-4 (2021) 1276–1313.
- *Generalized cusps in real projective manifolds: classification*  
(with D. Cooper and A. Leitner), J. Topol. 13-4 (2020) 1455–1496
- *Properly convex bending of hyperbolic manifolds*  
(with L. Marquis), Groups, Geom. & Dynam 14 (2020), no. 2, 653-688.
- *Convex projective structures on non-hyperbolic 3-manifolds*  
(with J. Danciger and G.S. Lee), Geom. & Topol 22 (2018) 1593–1646.
- *Constructing thin subgroups commensurable with the figure-eight knot group*  
(with D.D. Long), Algebr. Geom. Topol. 15 (2015) no. 5, 3011-3024.
- *Finite volume properly convex deformations of the figure-eight knot*,  
Geom. Dedicata 178 (2015), 49-73.
- *Deformations of non-compact, projective manifolds*,  
Algebr. Geom. Topol. 14 (2014), no. 5, 2595-2625.

### In Preparation

- *Convex projective Dehn filling*  
(with J. Danciger, G.S. Lee, and L. Marquis).

### Selected Talks

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|--------------------------------------------------------------------------------------------------------------------------|----------|
| • <i>Parseval Frames on Vector Bundles</i><br>AMS Southeast Sectional Meeting,                                           | Mar 2026 |
| • <i>Parseval frames on vector bundles</i><br>Codes and Expansions (CodEx) Seminar, Online seminar                       | Feb 2024 |
| • <i>A convex projective Dehn filling theorem</i><br>AMS Southwest Sectional Meeting, GA Tech                            | Mar 2023 |
| • <i>Complex Projective Structures on Surfaces</i><br>University of Florida, Colloquium                                  | Feb 2022 |
| • <i>Properly Convex Dehn Fillings</i><br>Virginia Geometry Seminar                                                      | Oct 2021 |
| • <i>Triangulations in Low Dimensional Geometry and Topology</i><br>Cal Poly San Luis Obispo, Colloquium                 | Feb 2021 |
| • <i>Gluing Equations for Projective Structures on 3-manifolds</i><br>Oklahoma State University, Topology Seminar        | Nov 2020 |
| • <i>Gluing Equations for Projective Structures on 3-manifolds</i><br>Trends in Low Dimensional Topology, Online Seminar | Jun 2020 |

- *Gluing Equations for Projective Structures on 3-manifolds\**  
Oklahoma State University, Topology Seminar Apr 2020
- *Gluing Equations for Projective Structures on 3-manifolds\**  
AMS Sectional Meeting, Special Session on Trends in Teichmüller Theory Mar 2020
- *Gluing Equations for Projective Structures on 3-manifolds*  
FRG Workshop, University of Michigan Dec 2019
- *Gluing Equations for Projective Structures on 3-manifolds*  
Geometry Seminar  
Stanford University Mar 2019
- *Thin Groups via Bending*  
Topology Seminar  
Rice University Feb 2019
- *Thin Groups from Projective Geometry*  
Thin Groups in Number Theory, Geometry and Topology  
Rice University May 2018
- *Convex Projective Structures in Low Dimensions*  
University of Georgia, Topology Seminar April 2018
- *Generalized Cusps in Convex Projective Manifolds*  
GEAR Retreat, Stanford University August 2017
- *Generalized Cusps in Convex Projective Manifolds*  
Dynamics on Character Varieties Workshop, University of Rennes I June 2017
- *Rank 1 Deformations of Non-cocompact Hyperbolic Lattices*  
University of Texas Topology Seminar March 2017
- *Geometric Structures on Manifolds*  
UF/FSU Topology and Dynamics Meeting  
University of Florida February 2017
- *Classification of Generalized Cusps*  
Joint Mathematics Meetings, Special Session on Group Actions and Geometric Structures,  
Atlanta, GA January 2017
- *Classification of Generalized Cusps*  
California Institute of Technology, Geometry and Topology Seminar April 2016
- *Geometric Structures on Manifolds*  
Mathematics Colloquium  
Florida State University January 2016
- *Gluing Convex Projective Manifolds*  
Higher Teichmüller theory and Higgs bundles: interactions and new trends  
Heidelberg November 2015

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\*Cancelled due to COVID-19 pandemic

- *Convex Projective Structures on Manifolds*  
A Mini Course at Workshop on Geometric Structures  
KIAS, Seoul, South Korea November 2014
- *Geometry of Cusps in Convex Projective Manifolds*  
Workshop on Geometry and Physics, Univeristy of Pittsburgh May 2014
- *Geometry of Cusps in Convex Projective Manifolds*  
RTG Workshop on Geometric Structures and Discrete Groups, UT Austin May 2014
- *A Geometric Description of  $(P)SL_4(\mathbb{R})$  Hitchin Representations*  
Workshop on Higher Teichmüller-Thurston Theory, Northport, Maine June 2013
- *Convex Projective Structures on Non-compact 3-manifolds*  
California Institute of Technology, Geometry and Topology Seminar May 2013
- *Convex Projective Deformations of the Figure-8 Knot Complement*  
AMS Special Session on Algebraic and Geometric Structures of 3-manifolds April 2013
- *Convex Projective Deformations of Non-Compact Manifolds*  
Arizona State University Differential Geometry Seminar January 2013
- *Convex Projective Deformations of Non-Compact Manifolds*  
Purdue University Geometric Analysis Seminar January 2013
- *Convex Projective Structures on 3-manifolds*  
University of California at Santa Barbara Topology Seminar October 2012
- *Twisted Cohomology and Deformation Theory*  
GEAR Junior Retreat, Urbana, Illinois July 2012

## Teaching Experience

### Florida State University

- MAS 3105 *Linear Algebra with Applications* Spring 2026
- MAC 2313 *Calculus III* Fall 2025
- MTG 5932, *Differential Geometry of Principal Bundles* Summer 2025
- MAC 2313 *Calculus III* Spring 2025
- MGF 3301 *Intro to Advanced Math* Fall 2024
- MTG 5932 *Lie groups* Summer 2024
- MAS 5932 *Arithmetic groups* Spring 2024
- MAA 4402 *Complex Variables* Summer 2023
- MAC 2313 *Calculus III* Spring 2023
- MTG 5327 *Topology II* Spring 2023
- MTG 4212, *College Geometry* Fall 2022
- MTG 5326 *Graduate Topology I* Fall 2022
- MTG 5932, *Differential Geometry of Principal Bundles* Spring 2022
- MAD 3105, *Discrete Math II* Spring 2022
- MAC 2313, *Calculus III* Fall 2021
- MTG 5932, *Geometric Structures on Manifolds* Spring 2021
- MAD 3105, *Discrete Math II* Fall 2020
- MAC 2312, *Calculus II* Spring 2020
- MAC 2313, *Calculus III* Fall 2019

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|-----------------------------------------|-------------|
| • <i>MAC 2312, Calculus II</i>          | Spring 2019 |
| • <i>MTG 5327, Graduate Topology II</i> | Spring 2019 |
| • <i>MAC 2313, Calculus III</i>         | Fall 2018   |
| • <i>MTG 5326, Graduate Topology I</i>  | Fall 2018   |
| • <i>MAP 2303, ODEs</i>                 | Spring 2018 |
| • <i>MAC 2312, Calculus II</i>          | Spring 2017 |
| • <i>MAC 2313, Calculus III</i>         | Fall 2016   |

### University of California, Santa Barbara

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|--------------------------------------------------------|-------------|
| • <i>Math 260P, Projective Structures on Manifolds</i> | Spring 2016 |
| • <i>Math 108A, Introduction to Linear Algebra</i>     | Fall 2015   |
| • <i>Math 8, Transition to Higher Mathematics</i>      | Fall 2015   |
| • <i>Math 260P, The Casson Invariant</i>               | Spring 2015 |
| • <i>Math 6A, Vector Calculus</i>                      | Winter 2015 |
| • <i>Math 108A, Introduction to Linear Algebra</i>     | Fall 2014   |
| • <i>Math 260P, Representations of Surface Groups</i>  | Spring 2014 |
| • <i>Math 8, Transition to Higher Mathematics</i>      | Winter 2014 |
| • <i>Math 103, Introduction to Group Theory</i>        | Fall 2013   |

### Mentorship

#### Ph.D. Students

|                                             |         |
|---------------------------------------------|---------|
| • Jared Miller                              | 2024    |
| • Ferhat Karabatman (joint with T. Needham) | Current |

#### Faculty

|                                     |              |
|-------------------------------------|--------------|
| • Oishee Banerjee (teaching mentor) | 2023-Present |
| • Ali Kara (teaching mentor)        | 2024-Present |

#### Postdocs

|                                     |              |
|-------------------------------------|--------------|
| • Wenwen Li (teaching mentor)       | 2025-Present |
| • Brandon Doherty (teaching mentor) | 2023-Present |
| • Florian Stecker (supervisor)      | 2022-2024    |
| • Eunjung Noh (teaching mentor)     | 2021-2023    |
| • Ignat Soroko (teaching mentor)    | 2021-2022    |
| • Alex Casella (supervisor)         | 2018-2021    |

### Ph.D. Committee Member

- Pan Fang (advisor Tom Needham & Washington Mio)
- Mujtaba Ali (advisor Tom Needham)
- Theodore Weisman (advisor Jeffrey Danciger, University of Texas at Austin)
- Paul Eugenio (advisor Oskar Vafek, Physics)
- Anindya Chanda (advisor Sergio Fenley)
- Aamir Rasheed (advisor Wolfgang Heil)
- Jay Leach (advisor Kathleen Petersen)
- Opal Graham (advisor Phil Bowers)
- John Bergschneider (advisor Wolfgang Heil)
- Leona Sparaco (advisor Kathleen Petersen)

### Honors in the Major

- Zachary Levy (Committee member, Physics)
- Jessica Bellaire (supervisor) 2024

### Independent Study

- Joseph Gerbasio (2026, undergraduate, differential geometry)
- Jessica Bellaire (2022, undergraduate, mathematics of Euler disks)
- Daniel Hartman (2018, graduate student, geometric structures and characteristic classes)
- Jose Bousa (2017, undergraduate, projective geometry)

### Funding and Awards

#### Grants Awarded

- Committee on Faculty Research Support award, Council for Research and Creativity (COFRS), Florida State University, \$20,000, Summer 2022.
- Supplement to DMS 1709097, \$27,483, 2021-2023.
- NSF research grant DMS 1709097- *Geometry and topology of convex projective manifolds*, \$139,278, 2017-2021
- First Year Assistant Professor Award, Council for Research and Creativity, Florida State University, \$20,000, Summer 2017

### Conferences and Workshops

- AMS Special Session  
AMS Southeastern sectional meeting  
Florida State University Mar 2024
- UF/FSU Topology and Geometry meeting  
Florida State University Nov 2023
- AMS Special Session  
AMS Southeastern sectional meeting  
University of Florida Nov 2019
- UF/FSU Topology and Geometry meeting  
Florida State University Feb 2019
- GEAR network junior retreat  
Stanford University July 2017

### Departmental Service

- Executive Committee 2025-Present
- Director of Pure Mathematics 2023-Present
- Faculty Evaluation Committee 2023-Present
- Algebra Tenure Track Hiring Committee 2022-23
- Geometry/Topology Tenure Track Hiring Committee 2021-22
- Organized Job Application Workshop 2019
- Library Committee 2018-22
- Organizer (Topology seminar) 2018-2024

### University Service

- Road Scholar Committee 2025-Present
- Faculty Senator 2021-2026

### Professional Service

- Journal Referee
  - *Journal of Experimental Mathematics*
  - *Journal of Differential Geometry*
  - *Transactions of the AMS*
  - *Publications Mathématiques de l'IHES*
  - *International Mathematics Research Notices*
  - *Journal of the London Mathematical Society*
  - *Algebraic and Geometric Topology*
  - *Geometriae Dedicata*
  - *Pacific Journal of Mathematics*
  - *Proceedings of the AMS*
- Reviewer for Mathematical Reviews 2014-Present
- Judge for Capital Regional Science and Engineering Fair 2017, 2019, 2024

### Honors, Awards, and Fellowships

- Research Training Grant Postdoc, UC Santa Barbara 2013-2016
- Research Training Grant Fellowship, UT Austin 2011-2012
- Deborah Jackson Award recipient, Emory University 2007

### Professional Memberships and Associations

- Member of the American Mathematical Society (AMS) 2007-present.

*Last updated Feb 25, 2026*